Controle Fuzzy em Matlab

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Disponível em: https://github.com/LuisBaezN/FuzzyControl

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1. Controle de Estufa

Nesta solução, as variáveis consideradas são umidade (0U - 100U) e a temperatura (23 ° C - 27 ° C) para controlar a água (0 ° - 360 °) e o consumo do vento (0V – 5V).

Observe que o intervalo e a unith em cada variável são apenas representantes para demonstrar a função. Além disso, as funções de associação e as regras são planejadas apenas para visualizá -las.

Código Matlab

```
clear all ; close all;  clc

c1=0;
c2=0;

kH=0.01;

mHMB=0;
mHB=25;
mHM=50;
mHA=75;
mHMA=100;

xH=0:0.1:100;

subplot(2,1,1)

HMB=exp(-kH*(xH-mHMB).^2);
plot(xH,HMB)
hold on
HB=exp(-kH*(xH-mHB).^2);
plot(xH,HB)
HM=exp(-kH*(xH-mHM).^2);
```


1

```
plot(xH,HM)
HA=exp(-kH*(xH-mHA).^2);
plot(xH,HA)
HMA=exp(-kH*(xH-mHMA).^2);
plot(xH,HMA)

title('Input 1: Humidity')
grid on

kT=3;

mTMB=23;
mTB=24;
mTM=25;
mTA=26;
mTMA=27;

xT=23:0.1:27;

subplot(2,1,2)

TMB=exp(-kT*(xT-mTMB).^2);
plot(xT,TMB)
hold on
TB=exp(-kT*(xT-mTB).^2);
plot(xT,TB)
TM=exp(-kT*(xT-mTM).^2);
plot(xT,TM)
TA=exp(-kT*(xT-mTA).^2);
plot(xT,TA)
TMA=exp(-kT*(xT-mTMA).^2);
plot(xT,TMA)

title('Input 2: Temperature')
grid on

%disp('Output 1: Llave\n cC=0\n cUA=90\n cMA=180\n cCA=270\n cA=360')
cC=0;
cUA=90;
cMA=180;
cCA=270;
cA=360;
```


2

```
%disp('Ouput 2: Ventilador\n cMD=0\n cD=1.25\n cM=2.5\n cR=3.75\n cMR=5')
cMD=0;
cD=1.25;
cM=2.5;
cR=3.75;
cMR=5;

h=input('Humidity (0u - 100u): ');
t=input('Temperature (23°C - 27°C): ');

hMB=exp(-kH*(h-mHMB).^2);
hB=exp(-kH*(h-mHB).^2);
hM=exp(-kH*(h-mHM).^2);
hA=exp(-kH*(h-mHA).^2);
hMA=exp(-kH*(h-mHMA).^2);

tMB=exp(-kT*(t-mTMB).^2);
tB=exp(-kT*(t-mTB).^2);
tM=exp(-kT*(t-mTM).^2);
tA=exp(-kT*(t-mTA).^2);
tMA=exp(-kT*(t-mTMA).^2);

k1=min(hMB,tMB);
k2=min(hMB,tB);
k3=min(hMB,tM);
k4=min(hMB,tA);
k5=min(hMB,tMA);

k6=min(hB,tMB);
k7=min(hB,tB);
k8=min(hB,tM);
k9=min(hB,tA);
k10=min(hB,tMA);

k11=min(hM,tMB);
k12=min(hM,tB);
k13=min(hM,tM);
k14=min(hM,tA);
k15=min(hM,tMA);

k16=min(hA,tMB);
k17=min(hA,tB);
k18=min(hA,tM);
```


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```
k19=min(hA,tA);
k20=min(hA,tMA);

k21=min(hMA,tMB);
k22=min(hMA,tB);
k23=min(hMA,tM);
k24=min(hMA,tA);
k25=min(hMA,tMA);

y1=(k1*cMD+k2*cD+k3*cM+k4*cR+k5*cMR+k6*cMD+k7*cD+k8*cM+k9*cR+k10*cMR+
k11*cMD+k12*cD+k13*cM+k14*cR+k15*cMR+k16*cMD+k17*cD+k18*cM+k19*cR+k20
*cMR+k21*cMD+k22*cD+k23*cM+k24*cR+k25*cMR)/
(k1+k2+k3+k4+k5+k6+k7+k8+k9+k10+k11+k12+k13+k14+k15+k16+k17+k18+k19
+k20+k21+k22+k23+k24+k5);
y2=(k1*cA+k2*cA+k3*cA+k4*cA+k5*cA+k6*cCA+k7*cCA+k8*cCA+k9*cCA+k10*cCA+k
11*cMA+k12*cMA+k13*cMA+k14*cMA+k15*cMA+k16*cUA+k17*cUA+k18*cUA+k19*cU
A+k20*cUA+k21*cC+k22*cC+k23*cC+k24*cC+k25*cC)/
(k1+k2+k3+k4+k5+k6+k7+k8+k9+k10+k11+k12+k13+k14+k15+k16+k17+k18+k19
+k20+k21+k22+k23+k24+k5);

X1=['y1= ',num2str(y1),'V y2= ',num2str(y2),'°'];
disp(X1)

figure
hold on
subplot(25,2,1)
plot(xH,HMB)
grid on
subplot(25,2,2)
plot(xT,TMB)
grid on
subplot(25,2,3)
plot(xH,HMB)
grid on
subplot(25,2,4)
plot(xT,TB)
grid on
subplot(25,2,5)
plot(xH,HMB)
grid on
subplot(25,2,6)
plot(xT,TM)
grid on
```


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```
subplot(25,2,7)
plot(xH,HMB)
grid on
subplot(25,2,8)
plot(xT,TA)
grid on
subplot(25,2,9)
plot(xH,HMB)
grid on
subplot(25,2,10)
plot(xT,TMA)
grid on

hold on
subplot(25,2,11)
plot(xH,HB)
grid on
subplot(25,2,12)
plot(xT,TMB)
grid on
subplot(25,2,13)
plot(xH,HB)
grid on
subplot(25,2,14)
plot(xT,TB)
grid on
subplot(25,2,15)
plot(xH,HB)
grid on
subplot(25,2,16)
plot(xT,TM)
grid on
subplot(25,2,17)
plot(xH,HB)
grid on
subplot(25,2,18)
plot(xT,TA)
grid on
subplot(25,2,19)
plot(xH,HB)
grid on
subplot(25,2,20)
plot(xT,TMA)
```


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```
grid on

hold on
subplot(25,2,21)
plot(xH,HM)
grid on
subplot(25,2,22)
plot(xT,TMB)
grid on
subplot(25,2,23)
plot(xH,HM)
grid on
subplot(25,2,24)
plot(xT,TB)
grid on
subplot(25,2,25)
plot(xH,HM)
grid on
subplot(25,2,26)
plot(xT,TM)
grid on
subplot(25,2,27)
plot(xH,HM)
grid on
subplot(25,2,28)
plot(xT,TA)
grid on
subplot(25,2,29)
plot(xH,HM)
grid on
subplot(25,2,30)
plot(xT,TMA)
grid on

hold on
subplot(25,2,31)
plot(xH,HA)
grid on
subplot(25,2,32)
plot(xT,TMB)
grid on
subplot(25,2,33)
plot(xH,HA)
```


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```
grid on
subplot(25,2,34)
plot(xT,TB)
grid on
subplot(25,2,35)
plot(xH,HA)
grid on
subplot(25,2,36)
plot(xT,TM)
grid on
subplot(25,2,37)
plot(xH,HA)
grid on
subplot(25,2,38)
plot(xT,TA)
grid on
subplot(25,2,39)
plot(xH,HA)
grid on
subplot(25,2,40)
plot(xT,TMA)
grid on

hold on
subplot(25,2,41)
plot(xH,HMA)
grid on
subplot(25,2,42)
plot(xT,TMB)
grid on
subplot(25,2,43)
plot(xH,HMA)
grid on
subplot(25,2,44)
plot(xT,TB)
grid on
subplot(25,2,45)
plot(xH,HMA)
grid on
subplot(25,2,46)
plot(xT,TM)
grid on
subplot(25,2,47)
```


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```
plot(xH,HMA)
grid on
subplot(25,2,48)
plot(xT,TA)
grid on
subplot(25,2,49)
plot(xH,HMA)
grid on
subplot(25,2,50)
plot(xT,TMA)
grid on
```

2: Sistema de Balanceamento

Um sistema de equilíbrio pode ser construído com sistemas de controle, como um controlador PID, no entanto, um controle difuso também pode resolver o problema. As variáveis utilizadas são a distância (2cm - 32cm) que é um grau de entrada e rotação (75 ° - 105 °) como saída.

Este segundo exemplo usa um erro para equilibrar o objeto.

Código Matlab

```
% https://github.com/LuisBaezN/FuzzyControl/blob/main/b_system/b_system.m

clear all; close all; clc

% Gráficos de Entrada
k=0.05;

mC=2;
mM=17;
mL=32;

xD=2:0.1:32;

subplot(2,1,1)
DC=exp(-k*(xD-mC).^2);
plot(xD,DC)
hold on
```


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```
DM=exp(-k*(xD-mM).^2);
plot(xD,DM)
hold on
DL=exp(-k*(xD-mL).^2);
plot(xD,DL)
hold on
title('Distance')
grid on

% Gráficos de Erro
K=17;

mN=-15;
mCE=0;
mP=15;

xE=-15:0.1:15;

subplot(2,1,2)
EN=exp(-k*(xE-mN).^2);
plot(xE,EN)
hold on
EC=exp(-k*(xE-mCE).^2);
plot(xE,EC)
hold on
EP=exp(-k*(xE-mP).^2);
plot(xE,EP)
hold on
title('Error')
grid on

% Saída
ab=75;
c=90;
ar=105;

% Entrada
di=input('Distância Atual (2cm - 32cm): ');

DC=exp(-k*(di-mC).^2);
DM=exp(-k*(di-mM).^2);
DL=exp(-k*(di-mL).^2);

% Erro
e=K-di;

EN=exp(-k*(e-mN).^2);
```


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```
EC=exp(-k*(e-mCE).^2);
EP=exp(-k*(e-mP).^2);

K1=min(DC,EN);
K2=min(DC,EC);
K3=min(DM,EN);
K4=min(DM,EC);
K5=min(DM,EP);
K6=min(DL,EC);
K7=min(DL,EP);

a1=min(K1,DM);
a2=min(K2,DM);
a3=min(K3,DM);
a4=min(K4,DM);
a5=min(K5,DM);
a6=min(K6,DM);
a7=min(K7,DM);

Y1=(a1*ab+a2*ab+a3*c+a4*c+a5*c+a6*ar+a7*ar)/(a1+a2+a3+a4+a5+a6+a7);

X1=['Graus do Motor = ',num2str(Y1),'°'];
disp(X1)
```


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